$$S = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$p_{1,:} = \mathrm{softmax}(2, 1, 0)$$

$$\approx (0.665,\ 0.245,\ 0.090)$$

$$\ell_1 = -\ln p_{11} \approx 0.408$$